Cirrhotic coagulopathy management and ROTEM

Transplant fellow lecture
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Welcome to the liver transplant fellowship lecture series. For this presentation, the slides are available as a google doc with citations linked.

Managing bleeding is a central challenge during liver transplantation. However, unlike many surgeries with major bleeding, coagulation dysfunction one of the core elements of cirrhosis, and exuberant attempts to enhance coagulation can paradoxically worsen the situation.

PART I: What is cirrhotic coagulopathy?

PT/INR included in cirrhosis scoring, but not current *diagnosis* recommendations where platelets * and fibrosis * are more important.

Non-variceal GI bleeding most common significant bleed. * * (Varices more morbid) *

Bleeding after procedures uncommon (3% clinically significant, 1% major) *

Hematoma and pseudoaneurysm after cath more likely but uncommon. *

GI bleeding is extremely common in ESLD; however, ~ 90% attributed to portal hypertension. * *

Platelets *uncorrelated* to non-variceal bleeding risk, INR > 1.5 weakly so. * *

RCTs of rfVIIa normalized INR and showed no impact on variceal bleeding *

FFP use associated with worse variceal bleeding, death. *

Bleeding is a common feature in cirrhosis, though not usually the presenting symptom. Changes in coagulation function reflect the loss of hepatocellular function in advanced cirrhosis, with the PT entering the Child-Pugh score and MELD; however, they are a relatively late changes and not used in diagnostic scores.

Interestingly while variceal bleeding is the most salient bleeding because of its high morbidity, non-variceal GI bleeding is substantially more common. Bleeding related to procedures, while noticeably higher than other hospitalized patients is still uncommon.

And while it would be tempting to connect the bleeding that patients with cirrhosis experience to coagulation defects, it is much more strongly linked to the degree of portal hypertension. That line of evidence includes both observational studies and interventional trials of agents to normalize coagulation function.

What is cirrhotic coagulopathy?

Evidence of impaired coagulation

Prolonged bleeding in standardized tests correlated to degree of cirrhosis * *

Decompensated cirrhosis leads to massive coagulation defect *

Elevations in PT due to decreased f2, 5, 7, 9, 10, 11. *

INR not well standardized / reproducible in ESLD due to thomboplastin differences * *

Decreased fibrinogen * largely due to increased consumption *

Low platelet counts * correlated to bleeding times *

Mostly increased levels of fibrinolytic proteins, * t-PA increased. *

Hyperfibrinolysis detectable by multiple assays and clinically relevant * , plausibly related to extra-vascular consumption *

This isn't to say that coagulation defects aren't present. The oldest functional marker of impaired coagulation, the bleeding time with standardized blade injuries, increases with cirrhosis markers such as bilirubin. Additionally, in decompensated ESLD mucosal bleeding and delayed hemostasis is common.

As mentioned, INR is a relatively good marker of hepatocellular function due to the relatively short half lives of some of the factors with decreased production in hepatocellular failure. As a side note, many PT assays are optimized for warfarin dosing, and are not as reproducible for for the factor defects present in cirrhosis as we might like.

Additionally, in more advanced cirrhosis fibrinogen half life decreases, and fibrinogen levels as well. Interestingly the decreased platelet count with can be observed before INR changes is also correlated to bleeding times.

Finally, increases in fibrinolysis leads to weaker clots and premature clot disruption

What is cirrhotic coagulopathy?
Evidence of Procoagulant Dysregulation

DVT rate in ESLD higher (OR ~2) than similar hospitalized patients (more in men?) *

Increase in stroke rate HR ~ 1.7 *

HCC strongly associated with VTE *

Portal vein thrombus in ~ 15% of ESLD patients, more as disease advances *

Intraop PE and intracardiac thrombus around 1.5% * *

Postop DVT+PE probably ~ 5% * *

It's underappreciated by some clinicians that *compensated* cirrhosis is however a distinctly pro-thrombotic disease. There are much higher rates of DVT, stroke, and PE, as well as somewhat unique thrombosis like portal vein thrombus.

This remains true in the OR, where PE and intracardiac thrombus, while uncommon, are much more common than other major surgeries. Even after transplant, the rate of thrombotic events is high.

What is cirrhotic coagulopathy?
"Rebalanced" hemostasis - the procoagulant side

Low protein C, S + normal thrombomodulin -> normal / elevated thrombin generation *

Sometimes *increased* fibrinogen in sepsis * and especially in biliary disease *

Increased factor VIII *

Increased vWF, but seemingly decreased function. Overall more reactive with exogenous platelets. *

Variable changes in ADAMTS-13, mostly decreased (-> more large vWF multimers) *

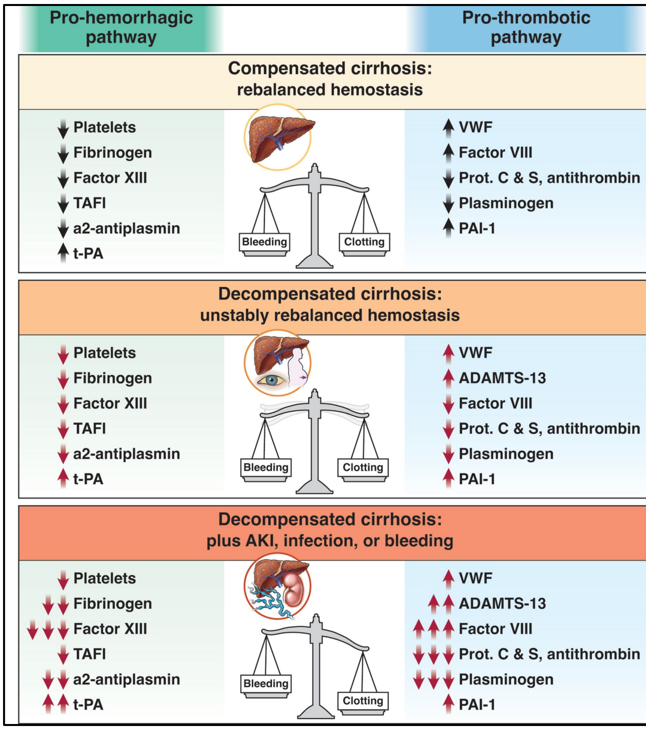
Decreased plasminogen, decreased anti-thrombin *

Fibrinolysis in transplant is extremely complicated, I recommend this review *

There are multiple mechanisms for this pro-thrombotic state, but the most immediately apparent is the decrease in proteins C and S, which leads to normal thrombin generation levels (although somewhat changed thrombin generation time courses) in most patients. Interestingly, although we highlighted hypofibrinogenemia as one of the hemostatic defects in ESLD, fibrinogen levels can be increased in acute infections, just as it is a normal acute phase reactant, and increased to very high levels in chronic biliary disease.

Factors *not derived from the liver* seem to contribute to the hypercoagulable state. Factor VIII and vWF are derived from the vascular endothelium and are up-regulated due to inflammation-like changes. vWF is a little complicated because the distribution of multimer sizes matters, and this seems to change over the course of ESLD. On the whole, the vWF that's present in compensated cirrhosis is as or more effective in engaging with platelets as part of primary hemostasis, and that partially is related to decreased ADAMTS-13, which normally cleaves some of the large vWF multimers.

Although we pointed to increased fibrinolysis as a coagulation defect, there are also several fibrinolytic factors that are deceased.



https://pubmed.ncbi.nlm.nih.gov/37121529

Thrombin generation effective with cryo + *5% plasma.* *

A platelet count of 50-100 maintains thrombin generation, but is slower. * *

This is a diagram summarizing the changes in hemostatic and fibrinolytic elements. To illustrate how narrow this balance can be, and how quickly it can change, the factors present in normal plasma and platelet counts can be enormously reduced and maintain thrombin generation.

PART 2: Treatments for coagulopathy

Do no harm

Strategies to "do less" seem to not worsen outcomes * *

Platelet transfusion observationally linked to early graft dysfunction, * mortality, * * * * lower reoperation *. May improve regeneration in LDLT * *

High-volume FFP robustly associated to poor outcomes * * * *

There have been many approaches to treating coagulopathy in cirrhosis and liver transplantation specifically. I'm going to quickly go over some of the major options that we use and think about, trying to note when they are given for specific deficiencies vs prophylaxis. There will also be some inferences from the same agents used in other settings.

Unfortunately, we don't often get to do trials of parachutes, and so many of the options that I'm going to discuss that have a rescue feel to them have especially low quality data or compare two options without a real control arm. For observational data, confounding by indication is a major concern. For example, RBC administration is highly linked to graft failure and death, but probably because of its administration in the context of bleeding. In most studies, it is extremely difficult to tease out the relationship between bleeding and multiple transfusion components given the limitations of data recording.

Before I go further I want to discuss some harms. There is observational data suggesting that most pro-coagulant treatments increase thrombotic events. However, I won't discuss these pro-thrombotic data much outside of trials, because large scale blood loss and worse cirrhosis confound their use and also cause thrombotic events, as we discussed earlier.

One theme that I won't explore in detail is that many groups have experience with implementing a bundle of a specific hemostasis treatment pathway, often with VET. A common finding is that they use many fewer procoagulant products with no change or slightly improved outcomes. This may be due to better targeting, or underappreciated harm, or overestimated benefit from products. Because these are bundles, we don't know the specific factors at play

Two important coagulopathy treatments are plasma and platelets, which have the major confounding problems that I mentioned. I found no trials of platelet or INR / clotting time targets. However, based on observational data adjusting for baseline platelet counts and RBC requirements, platelet transfusion is widely believed to be linked to worse outcomes related to sepsis and pulmonary complications despite being effective at reducing bleeding and reoperation. There is some intriguing human and animal data to suggest that the balance of risk may be different in living donor transplantation, with platelets somehow improving regeneration of the graft.

Similarly, high FFP use has been associated with poor outcomes in many studies. In addition to the confounding problem with bleeding, FFP use leading to high CVP seems to worsen intraoperative bleeding. CVP management and non-coagulation related bleeding management will hopefully be in a separate lecture.

Treatments for coagulopathy

PCC

Probably no different from FFP use (other than fluid volume) *, largest observational data with decreased bleeding intra and post * *

10-15 U/kg most common dose [1 U ~ 1 mL plasma]

Cardiac results favor PCC (~20 U/kg) at deeper levels of coagulopathy * * but not unambiguously *

Fibrinogen concentrate

Low quality data on fibrinogen targets, 150 mg/dL associated with bleeding * 200 with low platelets?

Dose (mg) = (Target − plasma concentration (mg/dL))/1.7 × body weight (kg). *

Small RCT of *preemptive* * fibrinogen with no effect on bleeding in OLT

11 vs 8 mm FIBTEM MCF target no differences in bleeding in OLT *

Most studies in *prophylaxis* (multiple surgery types) with no benefit *

Fibrinogen use *associated with* major thrombotic events after OLT *

Since I mentioned FFP, it's worth starting with PCC and fibrinogen concentrate. Although 4-factor PCC lacks many of the substances in plasma, there is little reason to administer it other than the lower fluid volume. There are no RCTs in liver transplant specifically of FFP vs PCC, with mixed observational data. In cardiac surgery there was a large RCT this year with reduced AKI and better hemostasis with PCC, but the other observational studies and smaller trials were not very favorable. It's probably the case that for very deep coagulopathy, it's impossible to feasibly administer the same amount of factors via plasma

We mentioned that many patients have low levels of fibrinogen, and can consume fibrinogen quickly due to fibrinolysis. Although there is not randomized quality data for target levels, 150 mg/dL has been associated in observational studies with more bleeding and is frequently used as a target minimum.

Fibrinogen concentrate is used heavily in many centers to achieve those targets because it can be rapidly available compared to thawed cryo from a blood bank. I will talk more about advantages and disadvantages versus cryo is 2 slides, but one thing to know about fibrinogen concentrate is that the amount of fibrinogen delivered is very consistent, which is useful if you are quite far from your target since you will likely get therapeutic in a single dose in a much smaller infused volume.

Unfortunately, the highest quality data comes from preemptive fibrinogen administration, where there is little evidence of benefit in liver transplantation, cardiac surgery, and trauma. The only trial that I found targeting a specific level in OLT is the deeply flawed TROMBOFIB trial. It used a FIBTEM target quite a bit higher than typical. So the 8mm MCF control arm there approximates a typical 150 mg/dL target. Unfortunately, the patient population that they recruited had a *median* FIBTEM-MCF of 10, so there was no difference in the amount of fibrinogen administered or levels achieved (average of 200 in both groups).

Treatments for coagulopathy

Cryoprecipitate

Cryo in cirrhosis outside of LT probably not helpful *

Supplemental cryo in trauma MTP probably not helpful *

Probably no difference in bleeding vs cryo (cardiac) * *

ROTEM use as an instrumental variable for cryo suggests small (OR 1.1/unit) increase in risk of major thromboembolism. * Also in small scale studies *

The other major options for treating low fibrinogen levels is cryo. Unfortunately the data to guide cryo administration is mostly outside of LT. Administration of cryo to critically ill patients with cirrhosis does not seem to decrease bleeding or improve survival, which tracks with our earlier discussion of bleeding being driven largely by other factors.

Supplemental cryo not targeting a fibrinogen level in trauma following empirical transfusion ratios doesn't seem to reduce blood loss. The only head-to-head trials of cryo and fibrinogen that I saw were in cardiac surgery, where there was no difference in blood loss.

I mentioned that protocols with treatment algorithms often show some benefit. These bundles with VET often induce a substantial increase in cryo or fibrinogen concentrate use, partially because they often under-estimate lab fibrinogen levels. Using these as an instrumental variable, which isn't totally valid because they induce other changes, gives one of the more valid seeming risk increases for thrombotic events.

Attribute	Fibrinogen concentrate	Cryoprecipitate
Fibrinogen concentration	Standard concentration of ~1 g per vial, after reconstitution becomes 1 g per 50 mL	Variable concentration of ~120–796 mg per 15 mL in each individual single donor unit
Viral screening and/or inactivation	Nucleic acid testing for HIV, hepatitis A, B, and C, and human parvovirus in donor plasma units Additional viral inactivation through precipitation/adsorption/pasteurization processes	Nucleic acid testing for HIV, hepatitis B and C, and other viruses[a]
Factor content	Purified fibrinogen	Fibrinogen and other coagulation factors including VWF, FVIII, FXIII, fibronectin, and platelet microparticles
Aspect of hemostasis that is treated	Secondary hemostasis by increasing substrate for thrombin	Primary hemostasis by increasing VWF and platelet microparticles Secondary hemostasis by increasing substrate for thrombin and FVIII activity (intrinsic tenase activity)
Acquisition time	Rapid reconstitution in minutes can be rapidly administered to the patient after reconstitution	Kept frozen at −20 °C and requires 30–45 min to thaw, once available can be rapidly administered to the patient
Shelf life after reconstitution or thawing	Shelf life is up to 24 h after reconstitution	Limited shelf life after thawing of 4–6 h; FVIII activity degrades relatively quickly, fibrinogen is more stable
Risk for transfusion adverse events	Negligible risk of alloimmunization, TACO, TRALI	Low, but present risk of allergic transfusion reactions, alloimmunization, and other transfusion adverse events (TACO or TRALI)
Total volume	50 mL diluent volume per 1 g vial	70–120 mL volume for 10 unit pool
Thrombotic risk in FIBRES study[24]	7.0% overall incidence 4.6% stroke or TIA incidence	9.6% overall incidence 5.0% stroke or TIA incidence
Acquisition cost	Acquisition cost of ~$1000 per 1 g in the United States, lower acquisition cost in Europe and Canada of ~$400–$500 per 1 g	Acquisition cost of ~$300–$400 per 5–6 unit pool in the United States Additional "hidden" costs include blood bank processing (~45 min to 1 h) and wastage, which increase the total cost

Cryo also contains vWF, ADAMTS-13,fibronectin, fVIII, **fXIII**

I mentioned that there is probably little difference in efficacy between cryo and fibrinogen concentrate, but it's important to be cognizant of some differences. As I mentioned, cryo has a somewhat variable fibrinogen concentration because it's derived from humans with variable fibrinogen concentrations. Cryo is a much larger volume of fluid, which may be relevant if you are correcting someone who is volume overloaded at the start of a case. Cryo contains a number of other substances, with fVIII and vWF being important in MTP scenarios, but these are typically normal to elevated in cirrhosis, so there is probably not much benefit from that addition. However, cryo also contains fXIII, fibrin stabilizing factor, which is often deficient in cirrhosis and may help treat through minor fibrinolysis.

There are a number of less relevant differences, for example the cost of fibrinogen concentrate is somewhat higher, although probably not much considering blood bank costs. Cryo used to be time consuming to obtain because of the thawing step, but currently our cryo is all pre-thawed and shelf stable for several days, meaning that we can bring it to the OR in anticipation.

Finally, Cryo is a blood product with the safety concerns for viral transmission inherent in that category.

Treatments for coagulopathy
Tranexamic Acid

Early RCTs (* *) favored TXA prophylaxis, but not at small doses *, did not report thrombosis. Overall small set of RCTs * . One conference-only-reported RCT *

Observational data in larger groups less favorable for efficacy, but no association to thrombosis * *

In liver *resection* one large (1245!) trial with NS but suggestive (OR 1.7 p = 0.08) link to thrombosis *, in broader general surgical populations mildly effective without obvious thrombosis risk * * - note the largest trial (POISE-3) suggests higher risk

No mortality benefit in bleeding varices *

Renally cleared! 1g + 10 mg/kg infusion enough for ROTEM endpoints *

TXA is controversial, but widely used in the US and internationally for OLT. TXA use has been tested both for treatment and prophylaxis, where is probably has a modest effect reducing blood loss. TXA for *treatment* of fibrinolysis has a much better established effect. The main question in its use is the increase in thrombosis risk. The prophylaxis trials did not report an increase in complications, but there was a large recent trial in liver resection with a non-significant but potentially clinically meaningful increase in thrombosis risk. Interestingly, in observational data TXA is both less efficacious and less risky. TXA use outside of liver transplant does not seem to meaningfully increase thrombosis risk.

One note is on dosing. For reversing VET diagnoses fibrinolysis, a substantially smaller dose than is used in orthopedics and obstetrics appears to be sufficient, and due to the frequently decreased renal clearance intraop additional dosing or infusion is probably unnecessary.

Treatments for coagulopathy

DDAVP

DDAVP promotes vWF and fVIII, and has effects on platelet granule release (at least in vitro improvement in platelet dependent thrombin generation) *

It has *minimal effect in cirrhosis* which tends to already up-regulate those factors. *

There is *minimal to no* in-vivo data to support use in uremic platelet dysfunction, but some data in other platelet-storage defects *

Many RCTs in multiple surgery types with and without platelet dysfunction with no benefit, probable harm in causing hypotension. Perhaps a small benefit in cardiac surgery. *

DDAVP probably useful in preventing / treating rise of sodium post-transplant *

DDAVP is uncommonly used at our center, but has a fair amount of history to it, and is occasionally requested by the surgical team in the context of uremic platelet dysfunction.

DDAVP has multiple effects, but predominantly does two things: release vWF and FVIII from vascular endothelium and upregulates platelet granule activity at least in in-vitro tests. The former is probably not relevant in cirrhosis, where those endothelial factors are already upregulated. However, if you were resuscitating from significant coagulopathy with PCC, which unlike plasma does not contain those factors, you might consider it. The best data in cirrhosis suggests that DDAVP as expected does not meaningfully impact coagulation.

DDAVP's effect on platelet function was originally investigated in genetic defects in platelet granule storage, and migrated to being used for uremic dysfunction with really no supporting data. Unlike some of the other agents being talked about, it has been extensively studied in trials, and outside of cardiac surgery where the CPB machine degrades vWF large multimers, it seems to have no benefit regardless of platelet dysfunction.

Interestingly, DDAVP's *other function* of being a vasopressin analogue probably is useful for limiting the rise of postoperative sodium by causing the kidney to retain free water in cases of preop hyponatremia.

Treatments for coagulopathy

Recombinant fVIIa

Non-significant difference in RBC transfused in small (n=82, 20-80 mcg/kg, n=182 120 mcg/kg) RCT in OLT * * Small difference in avoidance of any transfusion. Many small non-randomized studies with modest benefit. 20 mcg/kg prophylaxis in OLT seemingly ineffective with 2/38 thrombotic deaths *

Similar findings is liver resection in cirrhosis *

Possibly effective in variceal bleeding (post-hoc subgroup CP-B+ with active bleed during endoscopy) with 2.5% arterial thrombus *

In all studies remarkably improves coagulation measures

Dosing all over the place: cardiac 10-90 mcg/kg! * Cardiac societies recommend 20-40 mcg/kg as salvage *

Increased venous and arterial thrombosis in other surgery types (with no clear benefit) *, but some data suggesting lower risk *

Recombinant activated factor VII, which you will often hear referred to by the brand name novo-7 here, is rarely used at our center. There was a burst of data about novo7 in the early 2000s, with two small trials showing small benefits. It was much more widely studies in variceal bleeding, in which it was ineffective despite the favorable effects on coagulation parameters.

Some of the dispute on the efficacy of novo7 has to do with dosing. As you can see, the two RCTs used very different doses, including what we would now think of as very high dose 120 mcg / kg, and some studies using more typical 20 mcg/kg doses. The cardiac world, which has continued to see use of novo7 has mostly settled on it remaining a salvage option when all other measures are exhausted and at a lower dose.

The impact of novo7 on thrombosis has also been disputed. The studies in liver transplant have not reported high-frequency thrombosis, but in synthesis of other surgery types the risk of arterial and venous complications has been found to be noticable.

PART 3: VISCOELASTIC TESTING

PT, PTT, TT -> time to initial fibrin strands in plasma

Key problem: only semi-functional

TT increased in cirrhosis, classically used to detect very low fibrinogen (<100 mg/dL). TT *increases* with high fibrinogen, warfarin, * unreliable in cirrhosis * seems to vary more with antithrombin and heparinoids than fibrinogen.

Clauss fibrinogen is *extremely similar* procedure that dilutes plasma and uses much higher levels of thrombin to reduce the influence of inhibitors. * Fibrin times are compared to a reference sample / curve. Ongoing thrombolysis / DIC can artificially reduce result (backwards inhibition). DTI still affects, high dose heparin. *

There is a hemochon-like device measuring flow in thrombin-lined capillary. *

Thrombin based assays -> not clinically used

Traditional assays add triggering components to plasma and either optically or mechanically detect fibrin strands. They are poorly sensitive to how much fibrinogen is present, platelet interactions in clot formation, and the strength of the clot formed. They often are sensitive to just one aspect of the clotting process and therefore do not integrate the many changes in the "rebalanced" picture that we talked about and do not reflect clinical bleeding.

There are two traditional tests that I want to talk a little bit about because outside of liver transplant they don't get thought about very much by anesthesiologists.

The thrombin time, which measures fibrin formation with the addition of thrombin to plasma thereby bypassing most of the coagulation process. The time to form strands is related to concentrations of fibrinogen and thrombin, which is now standardized. When fibrinogen is low, the reaction to form that detectable fibrin is slower. It is sensitive to abnormal fibrinogen, and is usually prolonged in cirrhosis with acquired low fibrinogen. However, for it to show much sensitivity for low fibrinogen the level must be quite low, often sited at < 100. It also is very sensitive to antithrombin levels, split fibrin products, and direct thrombin inhibitors and so is not very useful in guidance. They are still sent by default, and in theory you will see a result suggesting a sudden decrease more quickly than a clauss fibrinogen measurement.

The standard for fibrinogen assay to a method called the Clauss determination, which is essentially the same as the TT except with a higher quantity of thrombin and diluting the plasma by multiple ratios. The curve from the different ratios is compared to reference sample to determine the fibrinogen level. These steps are all automated in current practice. The dilution factor and high thrombin concentration tends to eliminate or strongly reduce the impact of thrombin inhibitors, and gets around the nonlinearity and threshold effect in the TT. Several of the factors that affect TT can still artificially reduce the result, especially split fibrin products from fibrinolysis. It's worth noting that this is still a *functional* assay and not a chemical one: only fibrinogen that is actually available and able to transform into fibrin that forms strands is measured.

Diagnosing / assessing coagulopathy
Is ROTEM / TEG just the better test?

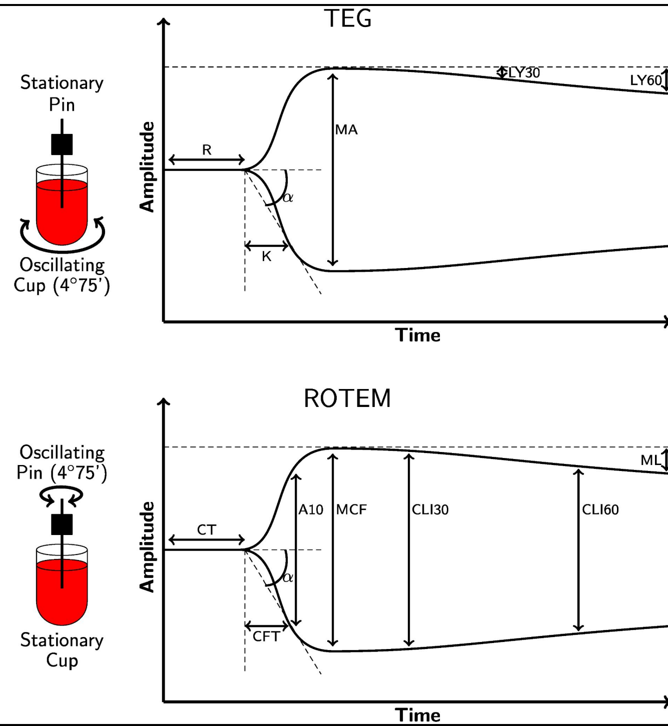
Desired:

- Measure of integrated / functional clotting
- Distinguish coagulopathy causes on the same assay
- Quick results
- Detect *hypercoagulability*

To overcome the issues surrounding classical tests, viso-elastic testing emerged as a more integrative measure. There are several things that we want out of an assay that we will look for as I talk about how they work and are interpreted.

First, we want actually integrative measures that strongly correlated to clinical bleeding. Second, we want to identify specific treatable problems on a single assay. Third, we want to do that in a reasonable amount of time. Finally, we'd like to be able to detect hypercoagulability to mitigate risk of thrombotic complications.

ROTEM vs TEG:
What are they

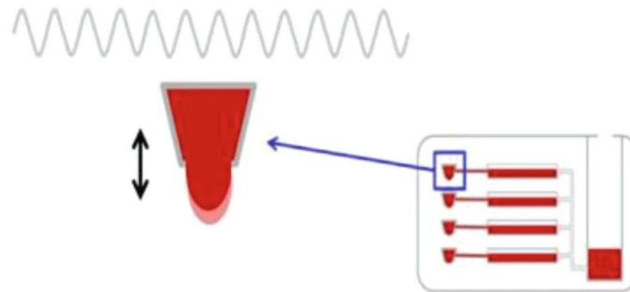


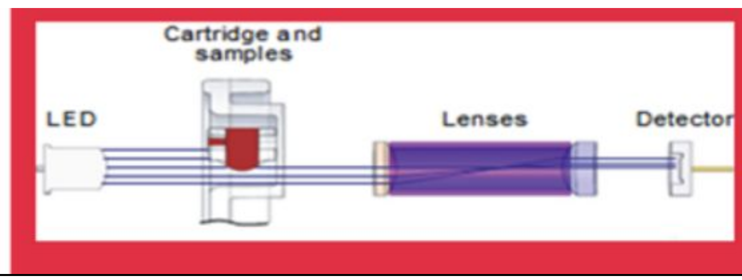
The mechanics of viscoelastic testing (which is the category including rotem, teg, and some competitors) are relatively simple. Rather than measuring fibrin formation in plasma, VET measure the viscosity of whole blood. Usually the time profile of viscosity is presented for interpretation, with the derived indices shown on the diagram here. Mechanically, you can use platelet rich plasma, and you will see plasma VET is some research papers, although obviously at POC you wouldn't be able to prepare that.

The mechanics of the two main VET are slightly different. TEG (which stands for thromboelastogram) has a cup with whole blood and a freely rotating pin sticking down into the blood. As the cup is rotated, for a low viscosity fluid, negligible rotational force is exerted on the pin. However, as the blood clots, increasing amounts of force cause the pin to rotate back and forth. A small reflector on the pin bounces a light beam to either side, and that deflection angle is what is shown on the envelope diagram. The rotational thromboelastomety device swaps the source of rotation. A spring provides a known rotational force to the pin. The cup of blood is held fixed, and as the blood begins to clot it offers resistance and absorbs some of the rotational energy. The pin therefore moves less as the blood clots and a detector aligned with the neutral position of the pin has a higher average intensity of light returned.

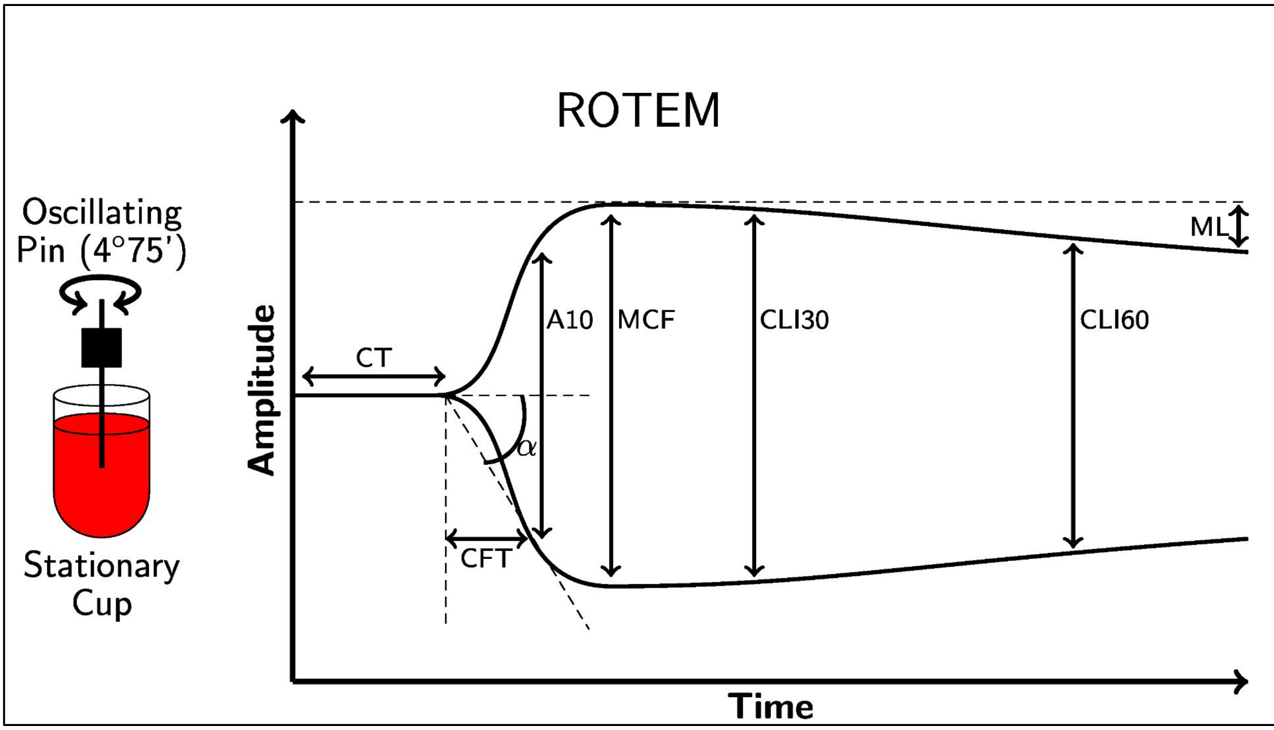
In most of what follows, I'm going to focus on ROTEM because it's used here, although an almost exchangeable set of points apply to TEG.

TEG 6s is resonance based Clot strength is determined using the resonance method, whereby the blood is exposed to a fixed vibration frequency range (20–500 Hz). Using light-emitting diode illumination, a detector measures the vertical motion of the blood meniscus.





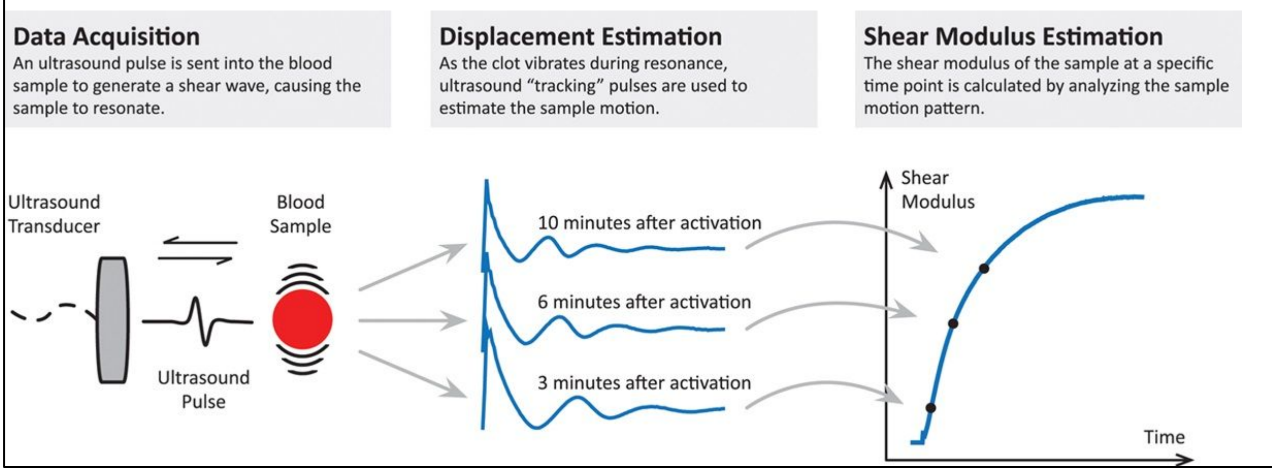
It's worth pointing out that several newer devices measure viscosity by alternative mechanisms. The approach that I just described turns out to be very sensitive to mechanical shock and vibration, and is pretty exacting the the quality control for the device. They also require manually pipetting the blood and reagents, which we'll talk about in a minute. The ROTEM sigma which we currently use is essentially the same with an automated cartridge for the sample processing. The TEG6s uses a completely different set of physics, and detects standing waves induced by fixed frequency vibration. The frequency that induces standing waves varies with the viscosity of the fluid.



I mentioned that these techniques show a time course of the blood viscosity. Interpreting the picture is useful, but derived numerical values allow reference values to be created, are more useful for research, and allow speedy interpretation in busy clinical settings. Many of the "strength" parameters are reported in "mm" reflecting the paper output the original devices created.

We will talk more about interpretation, but I want to describe these parameters so that some other aspects of the discussion will hopefully make a little sense. The clotting time is the time before any detectable change in the pin dynamics, and is analogous (but not exactly exchangeable with a PT or PTT. The maximum clot firmness measures overall clot strength, the A5 and A10 are clot firmness at 5 and 10 minutes used to approximate the MCF earlier. The clot formation time is the time between clot initiation and reaching a minimum strength of 20 mm. The alpha angle is the same information. The ratio of the maximum clot firmness to the values at 30 and 60 minutes are fibrinolysis indicators.

Quantra / sonorheometry



The quantra device is a related approach that uses ultrasound to first set a sample vibrating, then estimates the vibrational frequency based on reflected US pulses at different frequencies. That resonant frequency of the blood sample varies with the elasticity, which for gels like blood is closely related to viscosity.

Add some stuff to the cup

EXTEM, tissue thromboplastin (tissue factor).

FIBTEM, coagulation is activated as in EXTEM. cytochalasin D, the thrombocytes are blocked.

APTEM, coagulation is also activated as in EXTEM. addition of aprotinin or tranexamic acid in the reagent, fibrinolytic processes are inhibited in vitro.

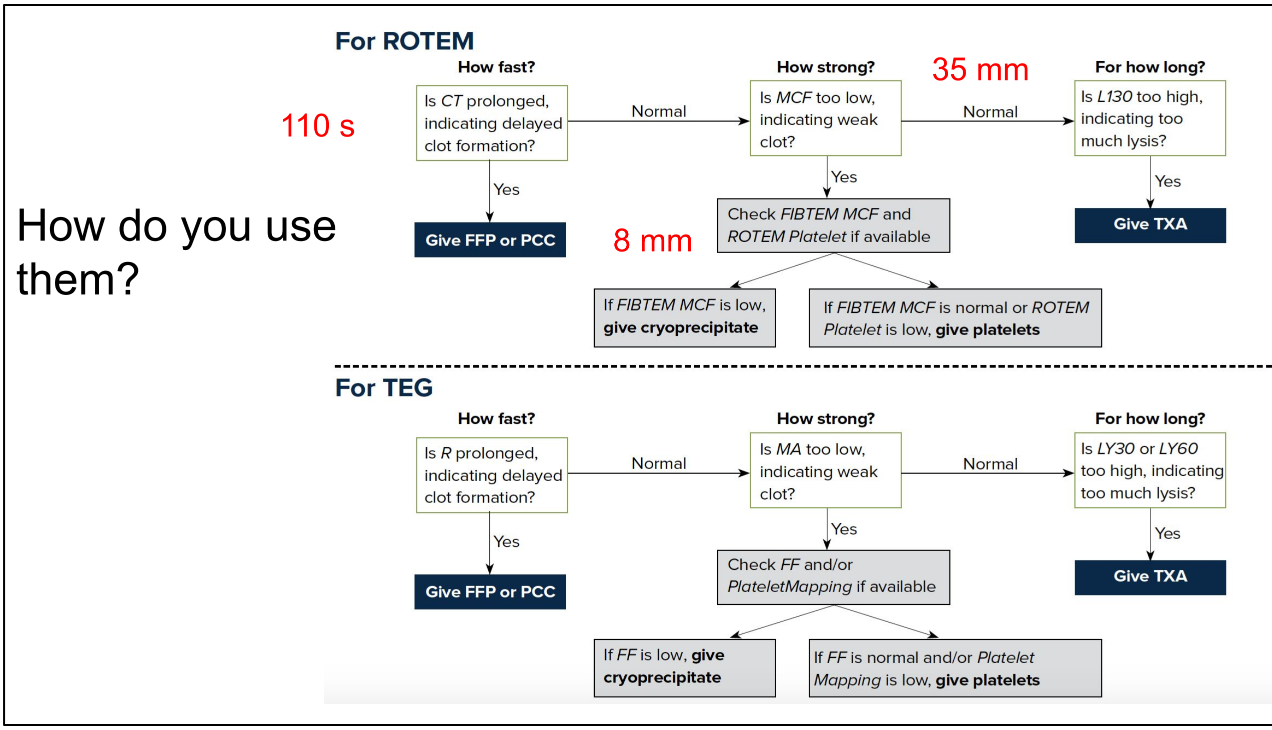
INTEM, contact phase (as in the aPTT and ACT). (Kaolin) The INTEM is sensitive for factor deficiencies of the intrinsic system (e.g. FVIII) and for the presence of heparin.

HEPTEM, coagulation is activated as in INTEM. The addition of heparinase in the reagent degrades heparin present in the sample and therefore allows the ROTEM® analysis in heparinised samples

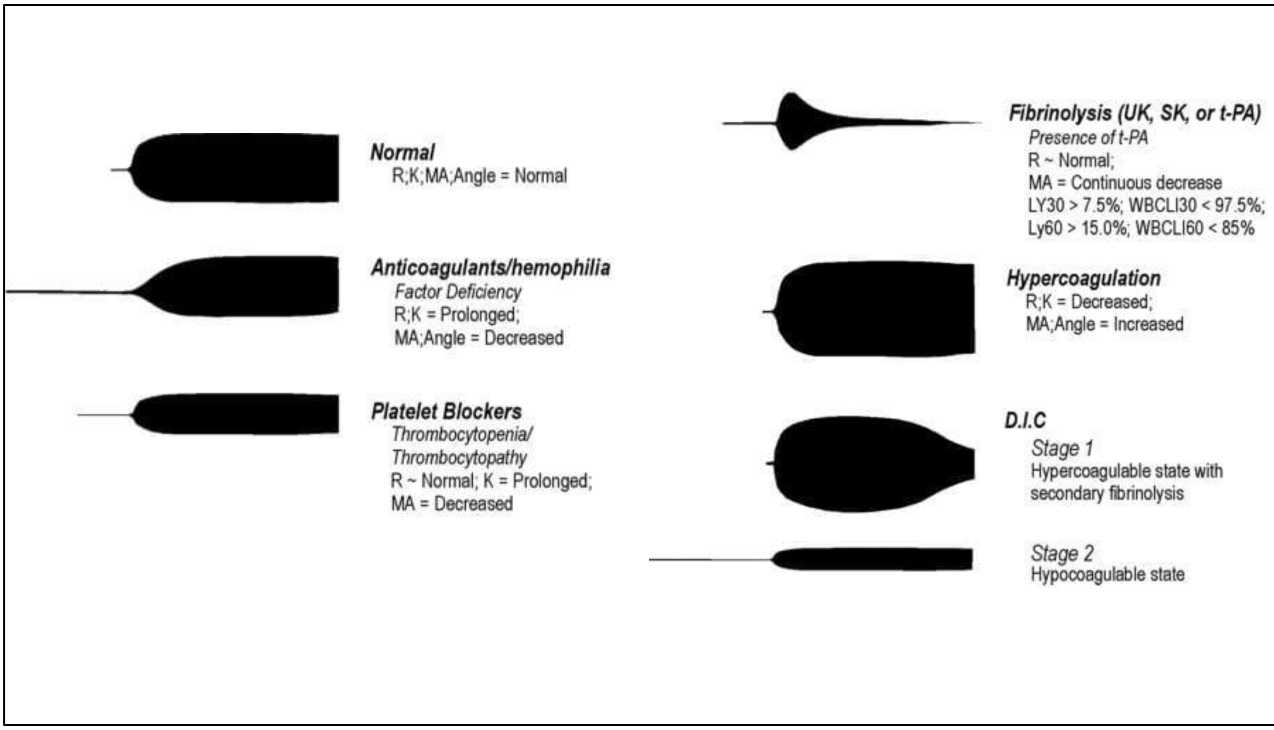
NATEM, add nothing

Ok, so we've figured out how to measure the viscosity or elasticity of blood over time. Just like with classical coagulation tests, you can get more information by exposing the usually citrated whole blood to reagents that emphasize different aspects of the clotting process. You will see some research with nothing added ("natem") as a kind of control when exploring different reagents, but it is not clinically used.

The main options are to add a substance activating the extrinsic pathway, like tissue factor in a PT, or the intrinsic pathway like Kaloin in a PTT. Additionally, it's common to add a platelet inhibitor to the EXTEM to isolate the contribution of fibrinogen to clot firmness. A fibrinolysis inhibitor like aprotinin can show when these agents would be effective in reducing lysis. A heparinase shows when heparin effects are present.



Integrating these measures into something that is essentially our current treatment pathway is pretty straightforward. Starting with an EXTEM, if the clotting time is long, about double the reference, there is probably a meaningful factor deficiency to correct. Next, if the MCF is weak, look at the fibtem. If them fibtem is 8mm or less, the fibrinogen level is probably around 150 or less, and cryo or fibrinogen should be used. If the lysis *at 30 minutes* is high, there is probably severe fibrinolysis that can be treated with low dose TXA. If the lysis only really develops later, like 45 minutes to 60 minutes, that is usually less relevant and a single pool of cryo may restore normal clotting, but it should be carefully watched. If the MCF is low and FIBTEM normal, this diagram mentioned he ROTEM platelet measurement, which we don't use, but a combination of platelet counts, other clinical information can suggest platelets. The same flowchart works in TEG.



Often people like these pictures instead of the indices, which are helpful if you have to read the diagram from across the room. This set of diagrams was created for TEG, but the ROTEM findings are essentially the same.

What is not included?

Not sensitive to aspirin, clopidogrel except very high doses

Not sensitive to LMWH

Anemia -> hypercoagulable

TEG / MCF inconsistently sensitive to FVIII, VWF

TEG FF is less good at eliminating platelet effects

FXIII important for all clot stability *

Flat line -> no clotting **OR** sample is clotted before the test started

There are of course many caveats to relying on ROTEM, not all of which have a good explanation. For example, although platelet function is included in the clot formation and clot firmness, it turns out that only very high doses of aspirin and clopidogrel affect the ROTEM. Similarly, LMWH's effect on Xa only minimally affects the typical ROTEM parameters. Anemia paradoxically increases the sample firmness because RBCs fill in some empty space in the clot network.

TEG is not as reliable detecting problems with vascular dependent factors, and the TEG version of the fibtem is generally less accurate and less able to eliminate the effect of platelets.

As mentioned earlier, there are factors like FXIII that contribute to clot stability that aren't fibrinolysis that have to be considered.

Finally, although it will rarely not be clinically obvious, a VET can produce a flat line when the sample is clotted before the test starts.

Fibtem assessment

Fibtem MCF moderately to strongly correlated to fibrinogen *

Notably poor agreement across devices and reagents *

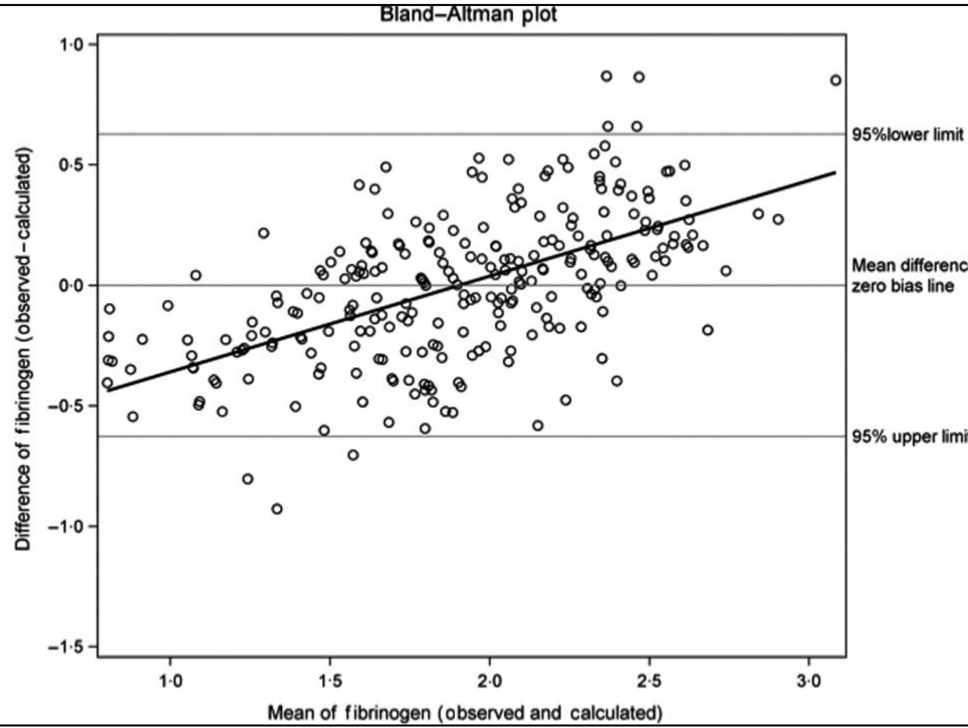
Because VET are *functional* administration of fibrinogen concentrate seems to *markedly worsen* the correlation to measured fibrinogen. *

Some LT specific studies with worse performance, particularly post-reperfusion *

A main rationale for VET is the ability to rapidly assess fibrinogen, often within 15 minutes, versus a Clauss taking 45 minutes to an hour in the lab. However, the correlation between Fibtem and Clauss varies from strong to only moderate in some studies. There is pretty poor standardization, and this correlation has varied over the generations of devices. As I mentioned the TEG version of fibrinogen doesn't seem to work as well. It also is apparently sensitive to endogenous versus fibrinogen concentrate, although this research was done at what we would consider supra-therapeutic fibrinogen levels, which may explain the degradation.

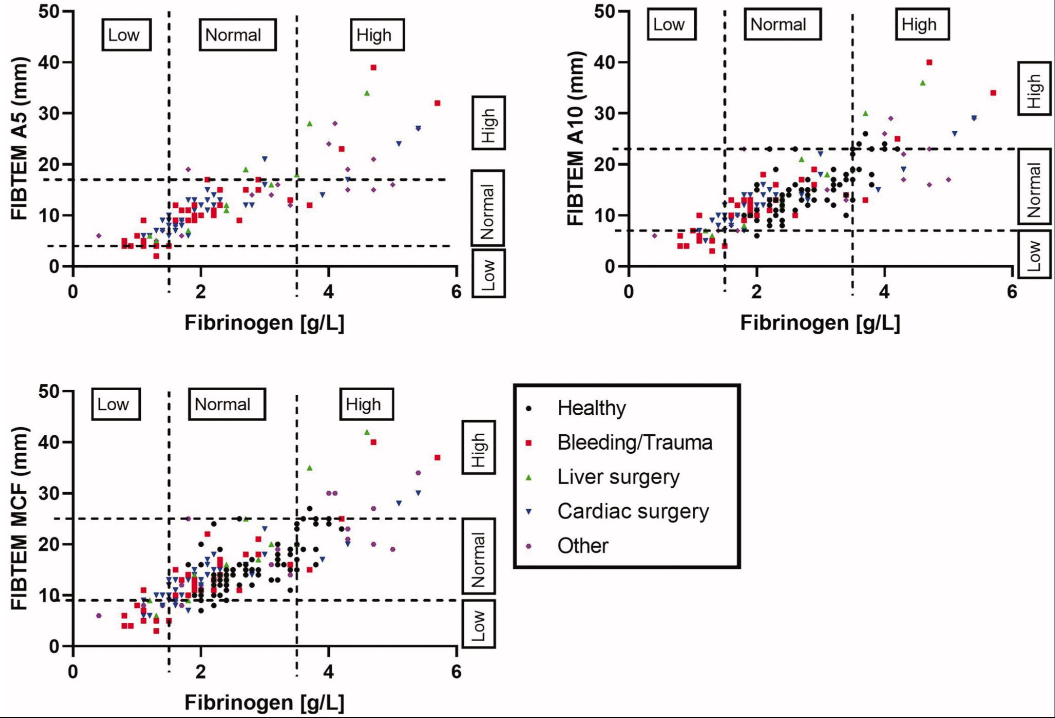
FIBTEM
Accuracy

*
–



This bland altman plot comes scandinavian liver transplants, and they have used a linear calibration between FIBTEM MCF and fibrinogen to be able to make a BA plot. Units are in g/L, so multiply by 100 for freedom units. As you can see, there is quite a bit of spread, about 50 mg/dL worth and some bias indicating nonlinearity in the MCF-fibrinogen relationship.

FIBTEM Accuracy



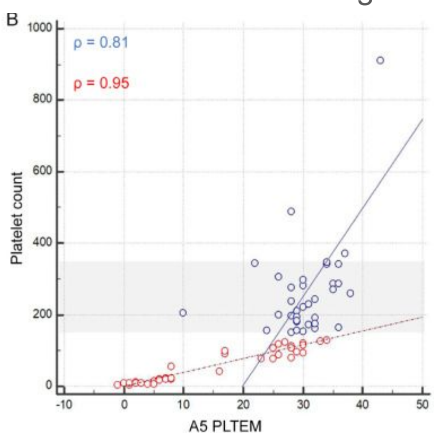
This paper considers multiple surgery types and the FIBTEM at 5 and 10 minutes, since those earlier markers are often used for rapid decision making. As you can see, in the 8-11 range we'd be interested in, there is again quite a bit of spread in the traditional fibrinogen measure.

Platelet estimation from ROTEM / TEG

This has never been accurate enough to label the device. (k is a lie) *

"Gold standard" platelet function is hard

A transformation should arguably be used *

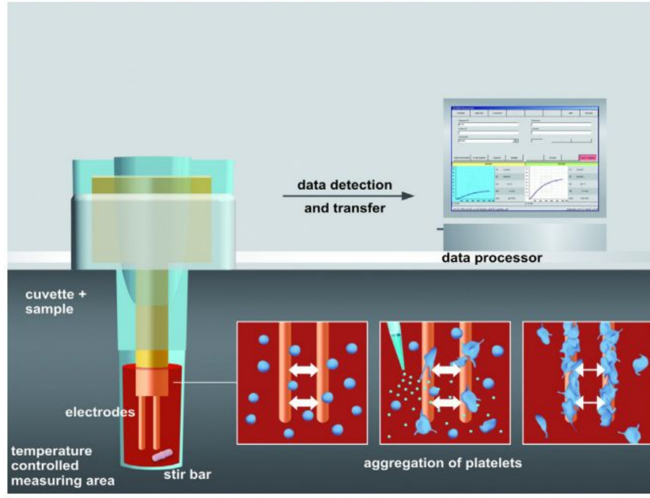


Clot elasticity	$CE = (100 \times A)/(100 - A)$ A represents clot amplitude
Maximum clot elasticity	$MCE = (100 \times MCF)/(100 - MCF)$ MCF stands for maximum clot firmness
Clot elasticity attributable to platelets (i.e., platelet component)	$CE_{platelet} = CE_{EXTEM} - CE_{FIBTEM}$ $MCE_{platelet} = MCE_{EXTEM} - MCE_{FIBTEM}$

I measured that there is a very natural attempt to assess platelet function with something like EXTEM-MCF weak and FIBTEM normal. Quantitatively this has been very difficult to make work despite multiple attempts, partially reflecting that platelet count is explicitly not what VET is measuring, and very few sources collect something like a PFA-200 for platelet function, which is still not a real gold standard.

The best attempt that I found reparameterized the MCF but there was no really data that it was better correlated to platelet count. In particular, in the very low range of platelets where we would normally treat vs cardiac and trauma, there is not enough supporting data.

The ROTEM® platelet module measures platelet aggregation respectively via electrical impedance based on impedance aggregrometry by Cardinal and Flower (1980). The ROTEM® platelet module is an impedance aggregometer intended for the assessment of platelet function in anticoagulated whole blood samples. It provides quantitative and qualitative information about the platelet aggregation by assessing the electrical impedance changes after platelet activation with different reagents.



The manufacturers of ROTEM and TEG acknowledge this weakness, and for the prior generation of devices included a platelet module which used a completely different approach. These devices measured the impedance of a sample after adding reagents, with increased as platelets adhered to the electrodes. They released validation data for identifying effects of aspirin and clopidogrel, similar to verifyNOW, but there is very little human data using ROTEM platelet. Currently, we rely on lab or POC platelet counts rather than a functional assay. As best I am aware, there is no rotem sigma analog.

FULL LENGTH ARTICLE · Volume 117, Issue 6, P741-748, December 2016 · Open Archive Download Full Issue

Rotation thromboelastometry velocity curve predicts blood loss during liver transplantation

L.A. Tafur · P. Taura · A. Blasi · ... · G. Martinez-Palli · J. Balust · J.C. Garcia-Valdecasas... Show more

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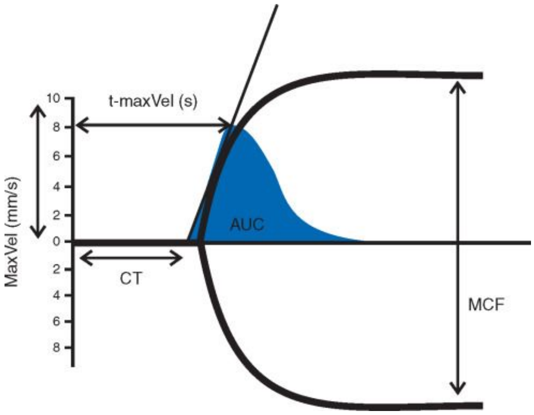
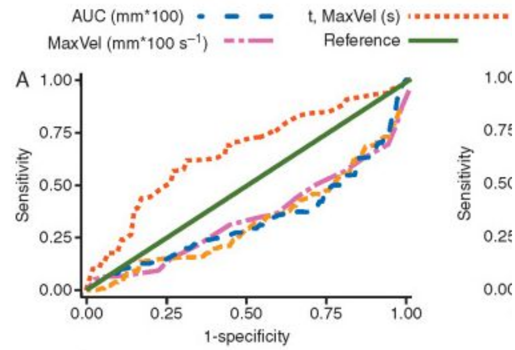




There are many references to the CFT and alpha angle as containing additional information regarding clotting; however, it turns out to be pretty minimal. The current ROTEM sigma devices display an *estimated* MCF before an actual maximum that incorporates the shape of the curve to predict where it is going to end up.

Can you do it … faster?

A10s widely used, highly correlated

A5s and alpha correlated but less tightly

Point-of-care Rotem Sigma saves overall time

Teg6s also somewhat faster

Comparative Study > Anesth Analg. 2019 Mar;128(3):414-423.
doi: 10.1213/ANE.0000000000003499.

Use of Earlier-Reported Rotational Thromboelastometry Parameters to Evaluate Clotting Status, Fibrinogen, and Platelet Activities in Postpartum Hemorrhage Compared to Surgery and Intensive Care Patients

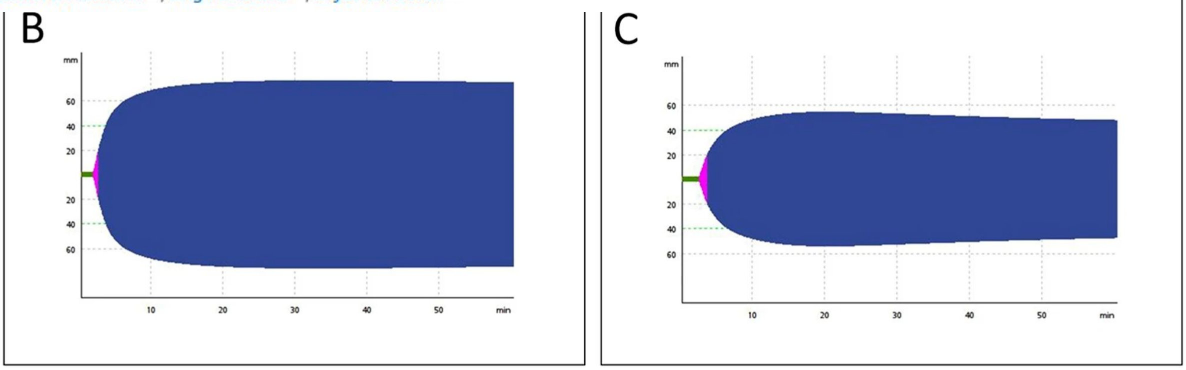
John G Toffaletti [1], Kelly A Buckner

Fibrinolysis is physiologic: "Fibrinolysis shutdown" in trauma

Multicenter Study > J Am Coll Surg. 2016 Apr;222(4):347-55.
doi: 10.1016/j.jamcollsurg.2016.01.006. Epub 2016 Jan 22.

Acute Fibrinolysis Shutdown after Injury Occurs Frequently and Increases Mortality: A Multicenter Evaluation of 2,540 Severely Injured Patients

Hunter B Moore [1], Ernest E Moore [2], Ioannis N Liras [3], Eduardo Gonzalez [2], John A Harvin [3], John B Holcomb [3], Angela Sauaia [2], Bryan A Cotton [3]



One other cautionary tale about ROTEM is that a *little bit* of fibrinolysis is normal, and the absence or below-normal lysis indicates additional clotting abnormalities and worse survival.

Diagnosing / assessing coagulopathy
Is ROTEM / TEG just the better test?

Smaller studies, but associated at least to bleeding in cirrhosis *

VET testing pre-procedure may reduce FFP + platelet use without additional bleeding *

No obvious decrease in bleeding when used to guide cirrhosis * * *

Rotem seems to detect variation in hypercoagulable components (fV, VWF, protein C+S) better *

fVIII and fXIII (fibrin stabilizing factor) strongly correlated to ROTEM MCF * *

TEG poorly correlated to classical parameters in ALF *

So, having described what VET are, is there evidence that this strategy works?

First, although the discrimination is not super high, EXTEM-CT seems to at least predict future bleeding in cirrhosis better than INR. It also is better able to detect the "rebalancing" effect in compensated cirrhosis with fewer false-positives.

Diagnosing / assessing coagulopathy

Does VET lead to better outcomes?

Recent studies continue to show decreased product use *

Probably more effective when paired with an explicit algorithm *

Now routinely used at most transplant centers in the US *

As I mentioned, many groups have included VET in algorithmic treatment pathways. Because these are inherently bundles and observational studies, it isn't clear what is driving changes, but they are well associated with decreased pro-coagulant product use and decreased PRBC requirement. They are now essentially standard of care.

Does it improve liver transplant outcomes? *

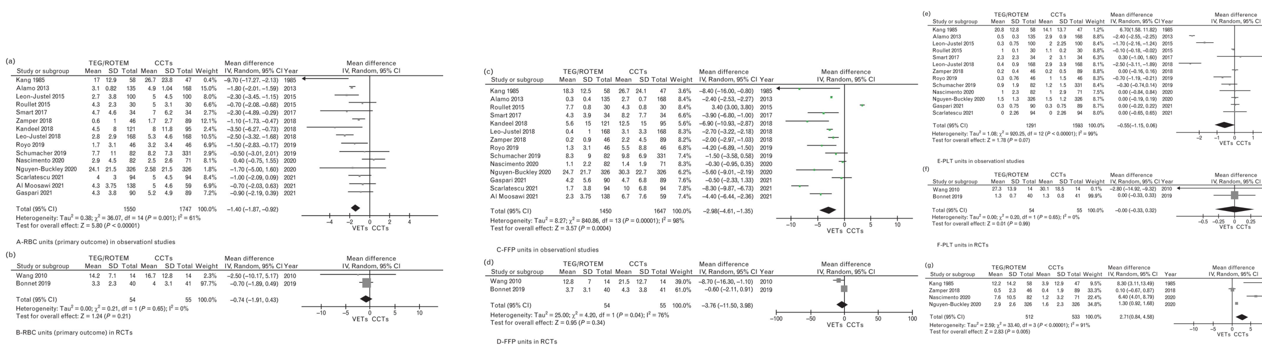
Meta-Analysis > Eur J Anaesthesiol. 2023 Jan 1;40(1):39-53.

doi: 10.1097/EJA.0000000000001780. Epub 2022 Nov 22.

Viscoelastic versus conventional coagulation tests to reduce blood product transfusion in patients undergoing liver transplantation: A systematic review and meta-analysis

Paola Aceto [1], Giovanni Punzo, Valeria Di Franco, Luciana Teofili, Rita Gaspari, Alfonso Wolfango Avolio, Filippo Del Tedesco, Domenico Posa, Carlo Lai, Liliana Sollazzi

Results: Seventeen studies (n = 5345 patients), 15 observational and two RCTs, were included in this review. There was a mean difference reduction in RBCs [mean difference: -1.40, 95% confidence interval (95% CI), -1.87 to -0.92; P < 0.001, I2 = 61%) and FFP units (mean difference: -2.98, 95% CI, -4.61 to -1.35; P = < 0.001; I2 = 98%) transfused in the VETs group compared with the CCTs one. A greater amount of cryoprecipitate was administered in the VETs group (mean difference: 2.71, 95% CI, 0.84 to 4.58; P = 0.005; I2 = 91%). There was no significant difference in the mean number of PLT units, mortality, hospital and ICU-LOS.



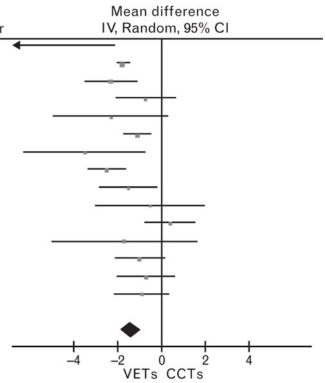
There however have been only 2 RCTs of VET in liver transplant. As you could imagine, there is a great deal of heterogeneity in the observational data, but the overall picture is decreased FFP and RBC use, increased cryo, and little effect on downstream outcomes.

Does it improve liver transplant outcomes?

(a)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Kang 1985	17	12.9	58	26.7	23.8	47	0.4%	−9.70 (−17.27. −2.13)	1985
Alamo 2013	3.1	0.82	135	4.9	1.04	168	15.0%	−1.80 (−2.01. −1.59)	2013
Leon-Justel 2015	2.7	3.8	100	5	4.5	100	8.0%	−2.30 (−3.45. −1.15)	2015
Roullet 2015	4.3	2.3	30	5	3.1	30	6.6%	−0.70 (−2.08. −0.68)	2015
Smart 2017	4.7	4.6	34	7	6.2	34	2.7%	−2.30 (−4.89. −0.29)	2017
Zamper 2018	0.6	1	46	1.7	2.7	89	12.1%	−1.10 (−1.73. −0.47)	2018
Kandeel 2018	4.5	8	121	8	11.8	95	2.4%	−3.50 (−6.27. −0.73)	2018
Leo-Justel 2018	2.8	2.9	168	5.3	4.6	168	10.5%	−2.50 (−3.32. −1.68)	2018
Royo 2019	1.7	3.1	46	3.2	3.4	46	6.9%	−1.50 (−2.83. −0.17)	2019
Schumacher 2019	7.7	11	82	8.2	7.3	331	2.9%	−0.50 (−3.01, 2.01)	2019
Nascimento 2020	2.9	4.5	82	2.5	2.6	71	8.1%	0.40 (−0.75, 1.55)	2020
Nguyen-Buckley 2020	24.1	21.5	326	2.58	21.5	326	1.8%	−1.70 (−5.00, 1.60)	2020
Scarlatescu 2021	4	3	94	5	4.5	94	8.4%	−1.00 (−2.09, 0.09)	2021
Al Moosawi 2021	4.3	3.75	138	5	4.6	59	6.9%	−0.70 (−2.03, 0.63)	2021
Gaspari 2021	4.3	3.8	90	5.2	4.9	89	7.2%	−0.90 (−2.19, 0.39)	2021
Total (95% CI)			1550			1747	100.0%	−1.40 (−1.87, −0.92)	

Heterogeneity: $Tau^2 = 0.38$; $\chi^2 = 36.07$, df = 14 ($P = 0.001$); $I^2 = 61\%$
Test for overall effect: Z = 5.80 ($P < 0.00001$)

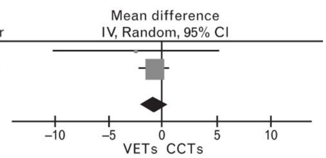


A-RBC units (primary outcome) in observationl studies

(b)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Wang 2010	14.2	7.1	14	16.7	12.8	14	2.3%	−2.50 (−10.17, 5.17)	2010
Bonnet 2019	3.3	2.3	40	4	3.1	41	97.7%	−0.70 (−1.89, 0.49)	2019
Total (95% CI)			54			55	100.0%	−0.74 (−1.91, 0.43)	

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 0.21$, df = 1 ($P = 0.65$); $I^2 = 0\%$
Test for overall effect: Z = 1.24 ($P = 0.21$)



B-RBC units (primary outcome) in RCTs

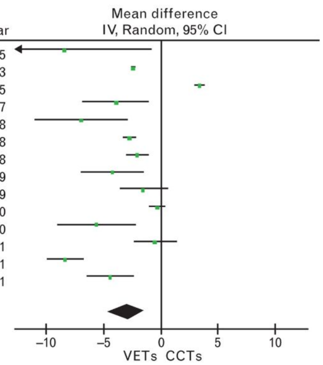
The reduction in RBC requirement has been pretty consistent , although smaller and NS in the randomized data

Does it improve liver transplant outcomes?

(c)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Kang 1985	18.3	12.5	58	26.7	24.1	47	3.6%	−8.40 (−16.00, −0.80)	1985
Alamo 2013	0.3	0.4	135	2.7	0.7	168	8.4%	−2.40 (−2.53, −2.27)	2013
Roullet 2015	7.7	0.8	30	4.3	0.8	30	8.4%	3.40 (3.00, 3.80)	2015
Smart 2017	4.3	3.9	34	8.2	7.7	34	6.6%	−3.90 (−6.80, −1.00)	2017
Kandeel 2018	5.6	15	121	12.5	15	95	5.6%	−6.90 (−10.93, −2.87)	2018
Leo-Justel 2018	0.4	1	168	3.1	3.3	168	8.3%	−2.70 (−3.22, −2.18)	2018
Zamper 2018	0.2	0.9	46	2.2	4.5	89	8.2%	−2.00 (−2.97, −1.03)	2018
Royo 2019	1.3	3.1	46	5.5	8.8	46	6.8%	−4.20 (−6.89, −1.50)	2019
Schumacher 2019	8.3	9	82	9.8	6.9	331	7.4%	−1.50 (−3.58, 0.58)	2019
Nascimento 2020	1.1	2.2	82	1.4	1.9	71	8.3%	−0.30 (−0.95, 0.35)	2020
Nguyen-Buckley 2020	24.7	21.7	326	30.3	22.7	326	6.2%	−5.60 (−9.01, −2.19)	2020
Gaspari 2021	4.2	5.6	90	4.7	6.8	89	7.6%	−0.50 (−2.33, 1.33)	2021
Scarlatescu 2021	1.7	3.8	94	10	6.8	94	7.8%	−8.30 (−9.87, −6.73)	2021
Al Moosawi 2021	2.3	3.75	138	6.7	7.6	59	7.4%	−4.40 (−6.44, −2.36)	2021
Total (95% CI)			1450			1647	100.0%	−2.98(−4.61, −1.35)	

Heterogeneity: $Tau^2 = 8.27$; $\chi^2 = 840.86$, df = 13 (P = 0.00001); $I^2 = 98\%$
Test for overall effect: Z = 3.57 (P = 0.0004)

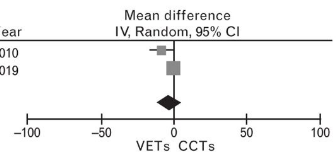


C-FFP units in observationl studies

(d)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Wang 2010	12.8	7	14	21.5	12.7	14	39.0%	−8.70 (−16.30, −1.10)	2010
Bonnet 2019	3.7	3.1	40	4.3	3.8	41	61.0%	−0.60 (−2.11, 0.91)	2019
Total (95% CI)			54			55	100.0%	−3.76 (−11.50, 3.98)	

Heterogeneity: $Tau^2 = 25.00$; $\chi^2 = 4.20$, df = 1 (P = 0.04); $I^2 = 76\%$
Test for overall effect: Z = 0.95 (P = 0.34)



D-FFP units in RCTs

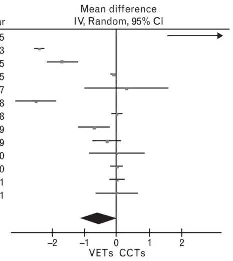
Same story with FFP

Does it improve liver transplant outcomes?

(e)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Kang 1985	20.8	12.8	58	14.1	13.7	47	1.2%	6.70(1.58. 11.82)	1985
Alamo 2013	0.5	0.3	135	2.9	0.9	168	8.8%	−2.40 (−2.55, −2.25)	2013
Leon-Justel 2015	0.3	0.75	100	2	2.25	100	8.4%	−1.70 (−2.16, −1.24)	2015
Roullet 2015	1	0.1	30	1.1	0.2	30	8.8%	−0.10 (−0.18, −0.02)	2015
Smart 2017	2.3	2.3	34	2	3.1	34	6.3%	0.30 (−1.00, 1.60)	2017
Leon-Justel 2018	0.4	0.9	168	2.9	3.9	168	8.1%	−2.50 (−3.11, −1.89)	2018
Zamper 2018	0.2	0.4	46	0.2	0.5	89	8.8%	0.00 (−0.16, 0.16)	2018
Royo 2019	0.3	0.76	46	1	1.5	46	8.3%	−0.70 (−1.19, −0.21)	2019
Schumacher 2019	0.9	1.9	82	1.2	1.5	331	8.4%	−0.30 (−0.74,0.14)	2019
Nascimento 2020	1	2.3	82	1	2.9	71	7.5%	0.00 (−0.84, 0.84)	2020
Nguyen-Buckley 2020	1.5	1.3	326	1.5	1.2	326	8.7%	0.00 (−0.19, 0.19)	2020
Gaspari 2021	0.3	0.75	90	0.3	0.75	89	8.7%	0.00 (−0.22, 0.22)	2021
Scarlatescu 2021	0	2.26	94	0	2.26	94	8.0%	0.00 (−0.65, 0.65)	2021
Total (95% CI)			1291			1593	100.0%	−0.55(−1.15, 0.06)	

Heterogeneity: $Tau^2 = 1.08$; $\chi^2 = 920.25$, df = 12 ($P < 0.00001$); $I^2 = 99\%$
Test for overall effect: Z = 1.78 ($P = 0.07$)

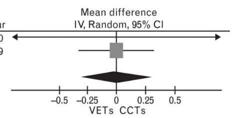


E-PLT units in observationl studies

(f)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Wang 2010	27.3	13.9	14	30.1	18.5	14	0.1%	−2.80 (−14.92, −9.32)	2010
Bonnet 2019	1.3	0.7	40	1.3	0.8	41	99.9%	0.00 (−0.33, 0.33)	2019
Total (95% CI)			54			55	100.0%	−0.00 (−0.33, 0.32)	

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 0.20$, df = 1 ($P = 0.65$); $I^2 = 0\%$
Test for overall effect: Z = 0.01 ($P = 0.99$)

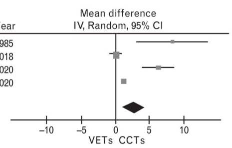


F-PLT units in RCTs

(g)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Kang 1985	12.2	14.2	58	3.9	12.9	47	9.5%	8.30 (3.11,13.49)	1985
Zamper 2018	0.5	2.3	46	0.4	1.9	89	33.3%	0.10 (−0.67, 0.87)	2018
Nascimento 2020	7.6	10.5	82	1.2	3.2	71	22.4%	6.40 (4.01, 8.79)	2020
Nguyen-Buckley 2020	2.9	2.6	326	1.6	2.3	326	34.8%	1.30 (0.92, 1.68)	2020
Total (95% CI)			512			533	100.0%	2.71(0.84, 4.58)	

Heterogeneity: $Tau^2 = 2.59$; $\chi^2 = 33.40$, df = 3 ($P < 0.00001$); $I^2 = 91\%$
Test for overall effect: Z = 2.83 ($P = 0.005$)



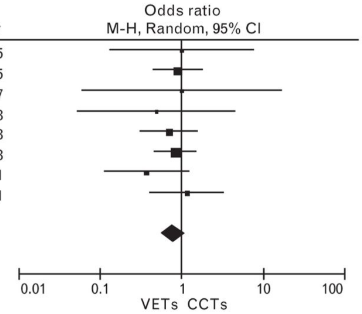
G-Cryoprecipitate units in observational studies

Platelets had the most heterogeneity. Many studies did not report cryo use, and not all centers really use cryo, but in the studies that did it increased. Similar increase in fibrinogen concentrate and PCC, lower TXA

(a)

Study or subgroup	TEG/ROTEM Events	TEG/ROTEM Total	CCTs Events	CCTs Total	Weight	Odds ratio M-H, Random, 95% CI	Year
Roullet 2015	2	30	2	30	2.8%	1.00 (0.13 to 7.60)	2015
Leon-Justel 2015	19	100	21	100	24.1%	0.88 (0.44 to 1.77)	2015
Smart 2017	1	34	1	34	1.5%	1.00 (0.06 to 16.67)	2017
Zamper 2018	1	45	4	89	2.4%	0.48 (0.05 to 4.45)	2018
Kandeel 2018	13	121	14	95	17.8%	0.70 (0.31 to 1.56)	2018
Leon-Justel 2018	24	168	28	168	33.0%	0.83 (0.46 to 1.51)	2018
Gaspari 2021	4	90	10	89	8.1%	0.37 (0.11 to 1.22)	2021
Scarlatescu 2021	8	94	7	94	10.4%	1.16 (0.40 to 3.33)	2021
Total (95% CI)		682		699	100.0%	0.79 (0.56 to 1.11)	
Total events	72		87				

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 2.56$, df = 7 ($P = 0.92$); $I^2 = 0\%$
Test for overall effect: Z = 1.37 ($P < 0.17$)

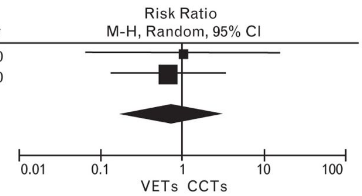


A-Mortality rate in observational studies

(b)

Study or subgroup	TEG/ROTEM Events	TEG/ROTEM Total	CCTs Events	CCTs Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Wang 2010	1	40	1	41	26.1%	1.02 (0.07 to 15.83)	2010
Bonnet 2019	2	14	3	14	73.9%	0.67 (0.13 to 3.40)	2010
Total (95% CI)		54		55	100.0%	0.75 (0.18 to 3.02)	
Total events	3		4				

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 0.07$, df = 1 ($P = 0.79$); $I^2 = 0\%$
Test for overall effect: Z = 0.41 ($P < 0.68$)



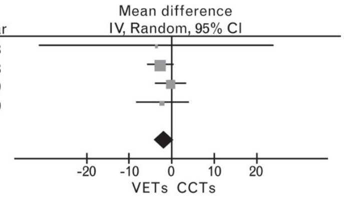
B-Mortality rate in RCTs

(c)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Zamper 2018	12.4	95	46	16.1	16.6	89	0.6%	−3.70 (−31.37 to 23.97)	2018
Leon-Justel 2018	38.2	14.4	168	40.9	13.8	168	51.4%	−2.70 (−5.72 to 0.32)	2018
Royo 2019	18.66	7.65	46	19	9.94	46	35.6%	−0.34 (−3.96 to 3.28)	2019
Nascimento 2020	18	16.3	82	20.3	21.5	71	12.5%	−2.30 (−8.42 to 3.82)	2020
Total (95% CI)			342			374	100.0%	−1.82 (−3.98 to −0.34)	

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 1.01$, df = 3 ($P = 0.80$); $I^2 = 0\%$

Test for overall effect: Z = 1.65 ($P < 0.10$)



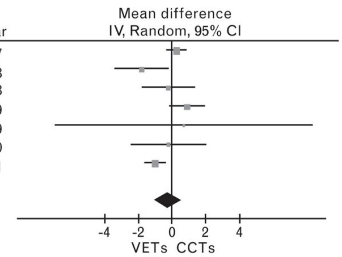
C-Hospital LOS for observational studies

(d)

Study or subgroup	TEG/ROTEM Mean	SD	Total	CCTs Mean	SD	Total	Weight	Mean difference IV, Random, 95% CI	Year
Smart 2017	2.6	0.77	34	2.33	1.54	34	24.1%	0.27 (−0.31 to 0.85)	2017
Leon-Justel 2018	8.4	7.5	168	10.2	7.6	168	12.3%	−1.80 (−3.41 to −0.19)	2018
Zamper 2018	3.4	4.3	46	3.6	4.6	89	12.7%	−0.20 (−1.77 to −1.37)	2018
Royo 2019	5	3.06	46	4.1	1.9	46	18.3%	0.90 (−0.14 to 1.94)	2019
Schumacher 2019	24.7	30.1	82	24	36.9	331	1.0%	0.70 (−6.93 to 8.33)	2019
Nascimento 2020	5	8.3	82	5.2	5.6	71	8.2%	−0.20 (−2.42 to 2.02)	2020
Scarlatescu 2021	5	2.26	94	6	2.26	94	23.3%	−1.00 (−1.65 to 0.35)	2021
Total (95% CI)			552			833	100.0%	−0.26 (−1.02 to −0.50)	

Heterogeneity: $Tau^2 = 0.53$; $\chi^2 = 16.53$, df = 6 ($P = 0.01$); $I^2 = 64\%$

Test for overall effect: Z = 0.67 ($P < 0.50$)



D-ICU-LOS for observational studies